ICNet: Cross-Modality Image Analysis for IC Localization in Printed Circuit Boards

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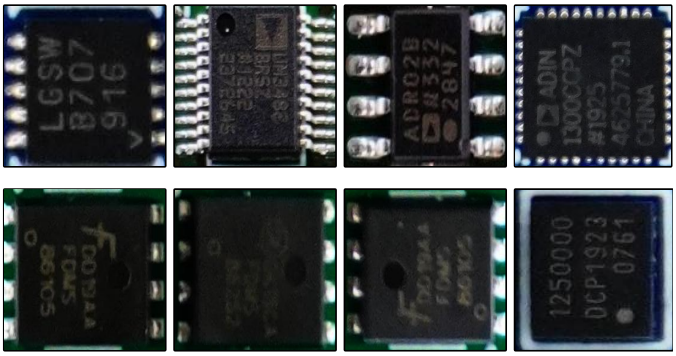
***Abstract*—Integrated circuits (ICs) are critical components on printed circuit boards (PCBs), and accurate IC localization is vital for automated PCB inspection and hardware security. However, existing optical-based methods often struggle to distinguish ICs from visually similar non-IC components. This paper proposes ICNet, a multi-stage IC detection framework that integrates optical detection with X-ray Computed Tomography (CT) verification. In the first stage, candidate ICs are detected from optical images with high recall to minimize missed detections. In the second stage, corresponding X-ray CT patches are derived through cross-modal processing and verified by a dual-branch classifier that fuses both semantic and structural features. Experimental results show that our proposed ICNet significantly reduces false positives and outperforms state-of-the-art methods, validating its effectiveness and reliability for real-world PCB inspection.**

***Index Terms*—Printed circuit boards (PCBs), object detection, multi-modal imaging, hardware security.**

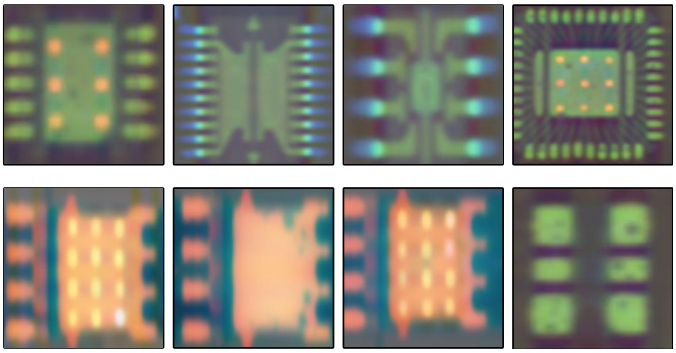
I. Introduction

Printed circuit boards (PCBs) are serving as the backbone of modern electronic system and are playing a pivotal role across a wide range of industries including critical infrastructure and national defense [1]–[3]. However, with the increasing reliance on outsourced manufacturing and the continued expansion of the global PCB market, the industry faces growing risks of malicious activities within the supply chain [4]. Consequently, there is a pressing need for automated and scalable PCB inspection techniques [5]–[7], particularly for verifying critical onboard components such as Integrated Circuits (ICs) [8].

Existing PCB inspection methods typically rely on optical imaging systems combined with computer vision techniques to perform tasks such as component detection, defect identification, and text recognition [9]–[11]. In [12], a deep convolutional neural network (CNN)-based object detector was employed for PCB component detection. In [13], a graph-based technique was developed to embed PCB image features for component identification. More recently, ECLAD-Net was introduced in [14] to detect resistors and capacitors based on optical PCB images, and PCBDet was proposed in [15] as a detection model optimized for edge devices. While these methods have demonstrated promising performance, they predominantly rely on optical imaging, which imposes inherent limitations in accurately distinguishing visually similar components [16]. As illustrated in Fig. 1(a), the optical images



(a) Optical images of ICs (top) and Non-ICs (bottom)



(b) X-ray CT images of ICs (top) and Non-ICs (bottom)

Fig. 1: Examples of (a) indistinguishable ICs and non-IC components under optical imaging, and (b) distinguishable ICs and non-IC components based on corresponding X-ray CT images where the ICs come with well-arranged pins.

of ICs (top row) and non-IC components (bottom row) exhibit nearly identical external appearances, posing significant challenges for the aforementioned optical-only inspection methods. To overcome this limitation, 3D X-ray Computed Tomography (CT) scanning provides a promising complementary modality that reveals the *internal structures* of PCB components. As shown in Fig. 1(b), it enables more reliable differentiation between ICs and non-IC components that appear visually similar in optical images but (internal) structurally different in X-ray CT images. While 3D X-ray CT scanning has been employed in PCB failure analysis tasks such as defect detection [17], [18], its potential for component identification based on internal structural characteristics remains largely

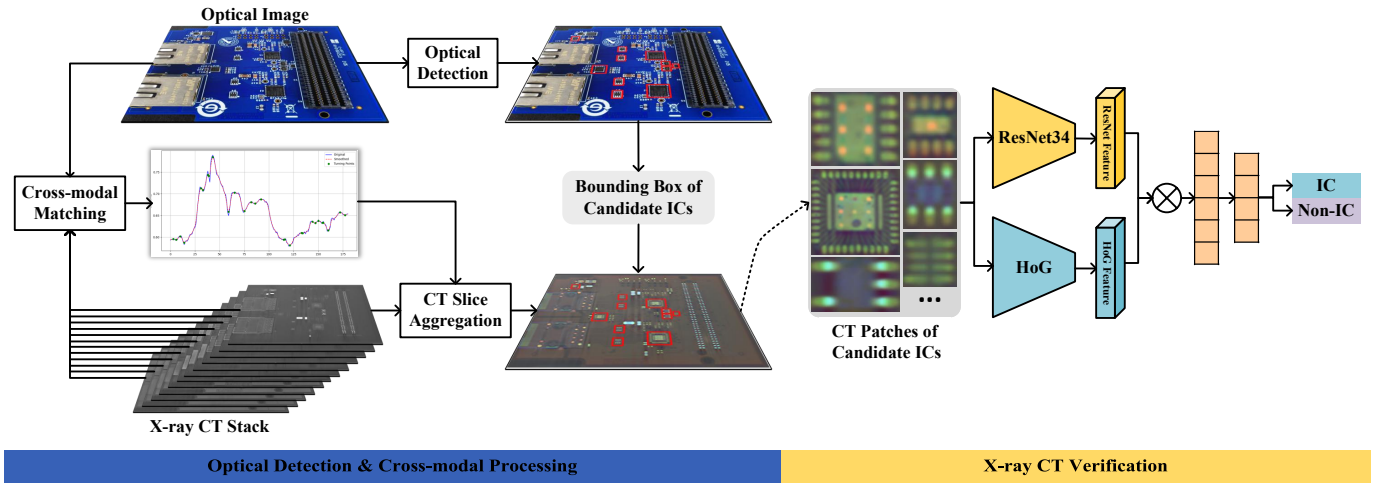


Fig. 2: Overview of our proposed ICNet framework for Integrated Circuit (IC) localization on printed circuit boards (PCBs). The Optical IC Detection stage applies YOLOv8 to optical images to generate candidate bounding boxes for potential ICs. From corresponding spatially-aligned X-ray images, X-ray patches of these candidate bounding boxes are extracted. In the X-ray IC Verification stage, each patch is represented by concatenating HOG descriptors and deep CNN features from a pretrained ResNet-34, which is then classified by a multilayer perceptron (MLP) to verify its class of IC or non-IC.

unexplored.

In this paper, we propose ICNet, a novel multi-stage IC detection framework that, for the first time, integrates both optical imaging and 3D X-ray CT scanning for IC localization. Unlike existing optical-only approaches, our proposed ICNet leverages complementary internal structural information from X-ray CT scanning to enhance detection reliability. Specifically, in the first stage, an optical-domain object detector exhaustively identifies candidate regions potentially containing ICs. In the second stage, the corresponding X-ray CT patches of these candidates are analyzed for further verification of ICs or non-IC components, using a combination of Histogram of Oriented Gradients (HOG) descriptors and learned CNN features. Experimental results demonstrate that our proposed ICNet significantly reduces false positives and improves detection performance in comparison to state-of-the-art methods.

The remainder of this paper is organized as follows. In Section II, we will delineate the overall architecture and key design modules of our proposed ICNet. We will then present the evaluation of our proposed ICNet with comparisons to several reported techniques in Section III. We will finally conclude this paper in Section IV.

II. Proposed Method

Fig. 2 presents the overall architecture of our proposed ICNet, which comprises two stages. In the first stage, candidate IC regions are generated using an optical-domain detector, and the corresponding X-ray CT patches are extracted through cross-modal image processing. In the second stage, these CT patches are further analyzed by a patch classifier to verify whether each candidate corresponds to an IC or a non-IC component. We now delineate these two stages in more details.

A. Optical Detection & Cross-modal Processing

For each PCB, optical imaging and 3D X-ray CT scanning will be performed to capture its surface appearance through an optical image and its internal structural details through an X-ray CT stack. The optical image and X-ray CT stack are first aligned spatially through image registration algorithm in [7]. The optical image is then processed by optical detection to generate bounding boxes $\mathcal{B} = \{B_i\}_{i=1}^{N}$ for N candidate ICs, which may include true positives (correctly detected ICs), false positives (non-IC components misidentified as ICs), and false negatives (missed ICs). In our approach, the optical detector is intensionally tuned (such as by adjusting the detection threshold) to minimize or ideally eliminate false negatives such that all potential ICs are captured for subsequent verification.

Meanwhile, each CT slice from the X-ray CT stack is matched to the optical image based on their Histogram of Oriented Gradients (HOG) features [19] to compute a similarity score. Let H_{opt} and H_i ($i \in 1, 2, 3, \ldots, m$) denote the HOG features extracted from the optical image I_{opt} and each X-ray CT slice X_i, respectively, where m is the total number of slices. The cosine similarity between H_{opt} and H_i is denoted as s_i, forming a similarity sequence $S = \{s_i\}_{i=1}^{m}$. A smoothed sequence $\hat{S}$ is obtained by applying a Gaussian filter $\mathcal{G}(\cdot)$, i.e., $\hat{S} = \mathcal{G}(S)$, as illustrated in the similarity score curve in Fig. 2. Based on $\hat{S}$, we determine a contiguous range of CT slices that correspond most closely to the optical image and collectively capture the most comprehensive structural information. To achieve this, we first identify the global maximum of the smoothed similarity curve as slice X_{max}, which represents the CT slice with highest structural similarity to the optical image and is regarded as the central slice of the contiguous range. If local minima (valleys in the similarity score curve) exists before and after the global maximum,

the nearest local minima (X_{bef} and X_{aft}) are used to define the starting slice as $X_s = X_{0.5(\text{max+bef})}$ and ending slice as $X_e = X_{0.5(\text{max+aft})}$. If such local minima does not exist, the search of X_s and X_e will start outward from the central slice until the magnitude of similarity change stabilizes over several consecutive slices. Slices within $[X_s, X_e]$ are considered to exhibit meaningful correspondence to the optical image, and any additional slices with similarity scores above 0.8 beyond this range are also included to ensure comprehensive coverage. The selected slices are then aggregated to form a three-channel X-ray CT image I_{ct} for subsequent processing.

Given the aggregated X-ray CT image I_{ct} and the set of bounding boxes $\mathcal{B}$ of candidate ICs, the CT patches P_j are cropped from I_{ct} according to each bounding box $B_j \in \mathcal{B}$, where $j \in \{1, 2, 3, \ldots, N\}$. These CT patches P_j are then fed into the second stage of our proposed ICNet for X-ray CT verification, where each candidate region is further classified as IC or non-IC.

B. X-ray CT Verification

In the X-ray CT verification stage of our proposed ICNet, the CT patches of candidate ICs are fed into a patch classifier to determine whether each candidate corresponds to an IC or a non-IC component. As shown in Fig. 2, the patch classifier comprises two complementary feature extraction branches, a ResNet34 branch that adaptively learns high-level semantic representations from the CT patches and a HOG branch that captures fine-grained structural details. Given CT patches $\{P_j\}_{j=1}^{N}$ of N candidate ICs, two feature vectors can be obtained from each P_j as the HOG feature vector $\mathbf{F}_j^h \in \mathbb{R}^{8100}$ and the ResNet34 feature vector $\mathbf{F}_j^r \in \mathbb{R}^{512}$. To enable effective feature fusion, both feature vectors are linearly projected into a common feature space and then concatenated as follows:

$$\mathcal{F}_j = W_h \mathbf{F}_j^h \oplus W_r \mathbf{F}_j^r, \tag{1}$$

where $W_h \in \mathbb{R}^{384\times 8100}$ and $W_r \in \mathbb{R}^{384\times 512}$ denote linear projection matrices for the HOG and ResNet34 feature vectors, respectively. The operator $\oplus$ represents feature concatenation, and $\mathcal{F}_j \in \mathbb{R}^{768}$ denotes the final fused feature vector.

The fused feature vector $\mathcal{F}_j$ is subsequently fed into a multilayer perceptron (MLP) to predict the probability of each candidate region being an IC. The classification process is defined as:

$$p(y_j|\mathcal{F}_j) = \mathrm{f}_{\mathrm{MLP}}(\mathcal{F}_j), \quad j \in \{1, 2, \ldots, N\} \tag{2}$$

where $\mathrm{f}_{\mathrm{MLP}}(\cdot)$ represents the learnt MLP decision function and $y_j \in \{0, 1\}$ denotes non-IC and IC labels, respectively.

At inference time, the X-ray CT verification process based on the patch classifier is applied to each candidate region to effectively suppress false positives (FPs) caused by optically similar non-IC components, while maintaining high recall for true ICs. By jointly leveraging high-level semantic information and low-level structural information, our proposed two-stage, cross-modality ICNet substantially enhances the reliability of component verification under challenging real-world conditions.

III. Experiment Results

In this section, we first describe the dataset used in our experiments, followed by the implementation details of our proposed ICNet and the state-of-the-art baseline methods. We then present the experimental results and comparative analysis, with discussions and future works provided at the end.

A. Dataset

Our dataset comprises paired optical images and X-ray CT stacks from 21 PCBs populated with surface-mounted components, including a total of 208 ICs. As described in Section II, the optical image and corresponding X-ray CT stack for each PCB are spatially aligned, and a three-channel X-ray CT image is generated through cross-modal processing. In total, the dataset contains 28 optical images and their corresponding X-ray CT images that include IC instances. The groundtruth bounding boxes of all ICs are annotated according to the respective Bill of Materials (BoM) for each PCB.

To facilitate model training and evaluation, the dataset is divided into training and testing subsets. The training set comprises 19 optical images (and their paired X-ray CT images), containing 155 IC instances, while the testing set includes the remaining images with 53 IC instances. This split ensures that the evaluation is conducted on unseen PCBs with a reasonable split ratio of ICs instances in training and testing subsets, thereby providing a fair assessment of the model's generalization capability.

For training the patch classifier in X-ray CT verification, we construct a balanced CT patches dataset derived from the aforementioned training subset. Specifically, CT patches for 155 ICs and 160 non-ICs are extracted from the 19 generated X-ray CT images in the training subset. To increase data diversity and mitigate overfitting, both classes are extensively augmented, resulting in 600 samples per class. This augmented dataset provides sufficient training samples for the patch classifier in the verification stage.

B. Implementations

In our proposed ICNet, we adopt YOLOv8 [22] as the optical detection model considering its favorable trade-off between accuracy and inference efficiency. The network parameters are initialized with pretrained weights from the COCO dataset [23]. All input images are resized such that the longer side is 2048 pixels while maintaining the original aspect ratio. The model is trained for 100 epochs using the default configuration, and the checkpoint achieving the highest mean Average Precision (mAP) on the validation split is selected for inference to produce candidate bounding boxes of potential ICs.

For the X-ray CT verification stage, the ResNet34 backbone is first trained from scratch for 100 epochs on our dataset to learn meaningful semantic representations of the CT patches. Subsequently, the HOG feature branch is integrated to enable the fusion of deep semantic and handcrafted structural features. The complete patch classifier is then trained for an additional 15 epochs. The checkpoint achieving the best classification

TABLE I: Performance comparison of our proposed ICNet with six benchmark models in terms of precision, recall, and F1-score. The first three rows report performances of EfficientDet [20], Faster R-CNN [21], and YOLOv8 [22] with only optical images. The fourth row reports performance of YOLOv8 with only X-ray CT images. The fifth and sixth rows report the performance of YOLOv8 using both optical images and X-ray CT images, but with either ResNet34 feature or HOG feature for X-ray CT verification. The last row reports the performance of our proposed ICNet. For each metric, the best and second-best performances are highlighted in **bold** and underline, respectively.

Index	Model	Modality	Precision↑	Recall↑	F1-score↑
1	EfficientDet [20]	Optical only	0.7692	0.5660	0.6522
2	Faster R-CNN [21]	Optical only	0.7500	0.7925	0.7706
3	YOLOv8 [22]	Optical only	0.7705	**0.8868**	0.8246
4	YOLOv8 [22]	X-ray CT only	0.7547	0.7547	0.7547
5	YOLOv8 + ResNet34 [22]	Optical & X-ray CT	0.8039	0.7736	0.7885
6	YOLOv8 + HOG [22]	Optical & X-ray CT	0.7925	0.7925	0.7925
7	Our Proposed ICNet	Optical & X-ray CT	**0.8545**	**0.8868**	**0.8704**

accuracy on the validation split is selected as the final model for IC verification.

For comparative evaluation with our proposed ICNet, we additionally implement EfficientDet [20], Faster R-CNN [21], and YOLOv8 with four different input and feature configurations, resulting in a total of six benchmark models for comparison.

C. Result Discussion & Future Works

Table I reports the IC detection performance of our proposed ICNet and the six benchmark models in terms of Precision, Recall, and F1-score. For each metric, the best performance is highlighted in bold and the second-best performance is underlined. Among the optical-only models (Rows 1–3), YOLOv8 achieves the highest recall indicating its strong sensitivity in detecting ICs and supporting our design choice of adopting YOLOv8 as the optical detector in ICNet. However, its precision remains significantly lower than our proposed ICNet, reflecting a substantial number of false positives when relying only on optical images. This demonstrates that complementary X-ray CT information is highly beneficial for rejecting false positives caused by visually similar non-IC components.

Comparing the third and fourth rows where YOLOv8 is applied using only optical or X-ray CT inputs, the X-ray CT configuration yields inferior results. Although X-ray CT captures internal structures, standalone X-ray CT detection tends to miss ICs due to the absence of strong surface appearances. This highlights the importance of pre-applying optical detection for exhaustively localizing candidate ICs. In the fifth and sixth rows, YOLOv8 + ResNet34 and YOLOv8 + HOG configurations demonstrate the complementary strengths of semantic and structural information. The ResNet34-based configuration achieves higher precision, while the HOG-based configuration achieves better recall. In contrast, Our proposed ICNet effectively combines both feature types, achieving the best precision, recall and F1-score among all.

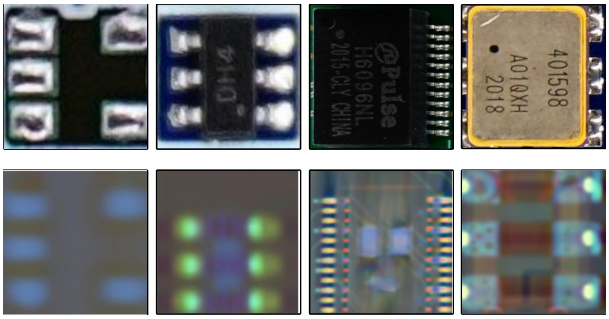


Fig. 3: Samples of difficult non-IC components.

In summary, the quantitative results confirm that our proposed ICNet substantially improves IC detection performance through its multi-stage and multi-modal framework, and validate that our proposed ICNet provides a practical and reliable solution for accurate IC detection in real-world PCB inspection. Despite these advantages of our proposed ICNet, a small number of false positives still remain in cases where non-IC components exhibit internal X-ray CT structures that closely resemble those of ICs. Fig. 3 shows the optical images (top row) and X-ray CT images of non-IC components which are difficult to distinguish from ICs in either image modality. This limitation highlights the inherent challenge of distinguishing non-IC components with similar surface and internal patterns as ICs. Future work could address this by incorporating multi-view X-ray CT scans to capture richer 3D structural information, albeit at the cost of increased computational complexity. Additionally, multimodal learning frameworks could be explored to enhance the cross-modal feature fusion between optical and X-ray CT images, enabling more robust detection across different PCB design or manufacturing variations.

IV. Conclusions

In this work, we presented ICNet, a multi-stage cross-modal framework that combines optical detection and X-ray CT verification for IC localization in PCB inspection. The first stage localizes candidate IC regions through optical detection, while the second stage leverages X-ray CT images with both structural information and semantic features to remove non-IC components. Experiment results demonstrated that our proposed ICNet significantly improves detection performance compared with state-of-the-art approaches.

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